

Meeting 5

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Date:	22 May 2025
Subject:	How Stable are Baselines: analysis on the Consistency of Reinforcement Learning Baselines
Mentor:	dr.ir. Á. Serra-Gomez
Project term:	February 2025 – August, 2025
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Outcome of previous meeting

- Rerun experiments with wrappers due to inconsistent use of the wrappers' internal normalization distribution whilst evaluating.
- Generate plots such that both unsmoothed and smoothed (only smoothed mean of repetitions) are displayed. Reproduction plots are plotted side by side for best comparison.
- Continue looking into SAC: run SAC on HalfCheetah-v4 for 3 million timesteps. Find what could have caused differences in performance.

Results during the last weeks

- Alignment of metrics, hyperparameters and wrappers of two baseline framework implementations must be done carefully: the process is very error prone
- PPO of Stable Baselines 3 and CleanRL differ in performance under the same hyperparameter configuration, implying implementational details affecting the consistency of the algorithm for a specific framework. Results under the original PPO hyperparameter configuration do match with the original results.
- SAC shows similar inconsistencies among Stable Baselines 3 and CleanRL, due to implementational (constant) details. These differences are small network differences and action picking methods, learning rate of the Adam optimizer for temperature optimization, and a (constant) weighted value loss.

Plans for next weeks

- I would like your opinion on extending this work to a third algorithm and/or a third framework.
- Start with structuring and writing the methodology.